MEBRIE AWOKE YIZNGAW

Full-Stack Developer | AI/ML Engineer

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Arat kilo, Addis Ababa, Ethiopia Mebrie Awoke Mebrie A.

Mebrie Awoke



Summary

Self-taught Software Developer and fourth-year Information Science student at Addis Ababa University, focused on Artificial Intelligence, Machine Learning, and Software Engineering. Experienced in building AI-powered applications, REST APIs, backend systems, and full-stack web and mobile solutions. Skilled in Python, FastAPI, Django, React, and modern AI technologies, with practical experience developing LLM-powered applications, RAG-based systems, and intelligent software solutions. Passionate about solving real-world problems through technology and developing reliable, impactful software.

Professional Experience

Innovation Intern, *Arada Subcity* Addis Ababa, Ethiopia

- Participate in software development and digital service initiatives within a public-sector innovation environment.
- Explore opportunities to improve administrative processes and service delivery through practical software solutions.
- Apply programming, API development, database design, and software engineering concepts to project development.
- Develop practical understanding of technology-driven public service improvement and digital transformation.

Website & Mobile App Developer paid Intern, *Orient PLC*

- Worked on web and mobile application development for an e-commerce platform.
- Implemented application features and integrated frontend and backend functionality.
- Worked with modern development tools and APIs.

Full-Stack Freelancer, *Upwork*

- Developed web applications and software solutions for client requirements.
- Worked across frontend and backend components.
- Used Git/GitHub and modern web-development technologies.

In addition to this, I have:

- Designed and developed personal software projects to solve real-world problems and strengthen my software engineering skills.
- Built and customized software solutions for small businesses, including clinics and shops.

Education

Addis Ababa University — Addis Ababa, Ethiopia
Bachelor of Science in Information Science 2023 – Present | Fourth-Year Student

Additional Learning

- Self-taught Software Development and Artificial Intelligence.
- Practical learning through personal projects, technical courses, and hands-on experimentation in AI/ML.

Technical Skill

Programming
Python, JavaScript, SQL, HTML5, CSS3, Java, C++

NLP & Generative AI
Transformers, LLMs, RAG, LangChain, Hugging Face, Semantic Search

Frontend
React, React Native

Tools
Git, GitHub, VS Code, Software Testing

AI & Machine Learning
PyTorch, scikit-learn, Deep Learning, CNNs, Transfer Learning, ML models like Supervised, Unsupervised and Reinforcement Learning

Backend
Flask, FastAPI, Django, REST APIs

Databases
MongoDB, SQL, ChromaDB

Projects

SmartFarm — AI-Integrated Agricultural Management System
Technologies: Python, HTML, CSS, JS, Machine Learning
- Develop an agricultural technology project integrating software engineering with AI-oriented functionality.
- Explore how intelligent systems can support agricultural activities and improve access to useful information.
- Design the application with a focus on practical usability and extensibility.

GitHub: SmartFarm

AI-Powered Chatbots — Adwa Victory, Medibot and Other RAG based Assistants
- Work on AI-driven chatbot solutions designed to provide users with accessible information through conversational interfaces.
- Apply backend development and AI integration concepts to connect conversational functionality with application services.
- Developed a medical chatbot using Retrieval-Augmented Generation (RAG) to answer questions from PDF documents.
- Integrated Hugging Face embeddings and ChromaDB for semantic search and relevant document retrieval.
- Used LangChain and Groq LLMs to generate context-aware responses.

Github: AdwaAI Medibot

RAG-Based Telegram Chatbot
Technologies: Python, Retrieval-Augmented Generation, Groq API, Telegram Bot API
- Develop a conversational application that integrates language-model capabilities into a Telegram interface.
- Explore retrieval-based approaches to providing relevant responses and building practical AI applications.
- Integrate external APIs and backend functionality to support chatbot interactions.

HireLens — AI-Powered Resume and Job Matching System
Technologies: Python, FastAPI, React
- Developed a job-fit web application that compares a candidate's resume and professional profile against a job description to assess job suitability.
- Generates a job-match percentage to help candidates understand how closely their qualifications align with a target role.
- Identifies matching skills and missing skills by analyzing resume content and job requirements.
- Provides personalized recommendations to help candidates identify skill gaps and improve their job applications.
- Applies text processing and AI-oriented analysis to support data-driven career guidance.

Github: HireLens

RoomPulse — AI-Powered Classroom & Library Monitoring
- Developed a real-time computer vision application for analyzing classroom and library environments.

- Implemented object detection and activity-level estimation with contextual AI feedback.
- **Technologies:** React Native, Expo, FastAPI, Python, YOLOv8n, OpenCV.

Github:RoomPulse

Interests

- Artificial Intelligence and Machine Learning
- Deep Learning and Computer Vision
- Natural Language Processing and Large Language Models
- Generative AI and AI Agents
- Applied AI Research and Intelligent Systems Development

Career Objective

To contribute to the development of practical, reliable, and impactful AI/ML solutions through model development, experimentation, data-driven problem-solving, and software integration while continuing to grow as an AI engineer and applied machine learning practitioner.